NHL Goal Prediction Research Protocol

Data contract temporal validation and reproducibility
Frozen first empirical study with technical appendices

Howard R. Silvey
Research edition 0.1.0
October 2, 2026

Original honours project supervised by Dr. Jason Nielsen
Carleton University School of Mathematics and Statistics
Original submission April 7, 2020

Independent 2026 research revision and extension. This edition is separate from the original university submission and does not imply institutional review or supervisor endorsement of the new analysis.

Primary sharing format PDF
Editable companion DOCX
Repository github.com/howardsilvey7/nhl-goal-prediction

Protocol purpose

This document specifies the first empirical study independently of its observed performance. The repository protocol was committed before development comparisons. The appendix records the actual frozen model choices without modifying the earlier search spaces.

The complete research programme is broader than this first release. This protocol defines what has been implemented and which advanced model families remain prospective. It supplies the conditions needed to interpret the thesis's numerical comparison.

Author: Howard R. Silvey. Frozen first empirical protocol: 2026-10-02. Research release: 0.1.0. This is a dated retrospective protocol, not a claim of external preregistration. Commit it before running development comparisons.

Research question

Using historical information from earlier source calendar dates, which practical statistical or machine-learning framework best predicts the paired regulation goal counts on subsequent NHL regular-season games and seasons?

The full project retains the broader pregame joint-distribution question in the transfer baseline. This initial experiment is a narrower, explicitly operational implementation. Calendar-date separation is verified; exact publication-time availability and archived historical source revisions are not verified.

Archival track

Preserve the 2020 PDF without editing. Local recovery has found that PDF but no original NHL R code or input table. Original numerical findings are source-verified where readable and are not computationally reproduced. Do not present a modern game-level ridge model as an exact reconstruction of the 2020 analysis. Archival source recovery remains open without blocking this separately labeled experiment.

Data and scoring

Use MoneyPuck's explicitly published all-team game download for non-commercial research and credit MoneyPuck.com. Raw data and row-level derivatives are not redistributed. Retain source URL, retrieval time and SHA256 locally and commit the provenance manifest. Do not scrape unlisted MoneyPuck pages.

Select season-start years 2016 through 2024, regular-season records only, and situation `all`. One canonical row per game joins exactly one HOME and one AWAY team record. Require score, opponent, date, season and duration agreement across the two records. Fail on duplicate roles or pairing inconsistencies.

Primary targets are regulation goals reconstructed from actual played goals and elapsed seconds. At 3600 seconds, use the played goals directly. Above 3600, subtract one actual overtime goal from the winning team; if played goals are tied, the game went to a shootout and no shootout marker is counted. Require duration between 3600 and 3900, and tied regulation scores after adjustment. Exclude incomplete/nonstandard records and retain an exclusion ledger. Check a stratified sample against official period-by-period scoring before modeling. This reconstruction and its source limitations must be disclosed in the report.

Information boundary

Forecast on the source game calendar day. All games with that date receive features before any same-date game updates the history. Use only strictly earlier dates. This is a conservative calendar-day design, not a fabricated UTC cutoff. No contemporaneous statistics, final standings, lineups, odds or goaltender claims are used. Goals, shots and attempts come from the preceding 20 included games, with 10 fixed prior games at 2.7 goals, 30 shots and 55 attempts per side. Historical league home/away rates start with 100 prior games at 2.8/2.6 goals. These priors are engineering choices fixed before comparisons, not estimated from holdout outcomes. Histories carry across seasons; rest is capped at 7 days. An expansion/renamed team starts from the pooled prior. ARI and UTA are distinct team identities in this first experiment, without an assumed franchise merge. Missing predictors cause a failure rather than silent zero filling.

All features derive from included regular-season games only. Playoff form is excluded. Shots/attempts describe prior completed games including their overtime, while goals are regulation-only. These mixed-exposure predictors are documented and may be refined in later versions. Historical source revisions are an inherent limitation even when date leakage is prevented.

Temporal validation

Foundation: 2016-17 through 2018-19. Outer development seasons: 2019-20, 2020-21, 2021-22, 2022-23 and 2023-24. For each outer season, use earlier seasons to fit; choose hyperparameters on the most recent earlier season, with only its preceding seasons used for the inner fit. Fit preprocessing separately in every inner and outer training window. Refit the chosen configuration on the full outer training window. Learned coefficients are fixed for the complete outer season, while lagged history features can update from earlier dates within it.

Final holdout: 2024-25. After development results, freeze the winning model family using game-weighted mean joint negative log likelihood over outer development games. Choose its final hyperparameters using 2023-24 with training through 2022-23, then refit through 2023-24. Freeze all comparison-model configurations the same way before one holdout evaluation. No holdout-dependent recalibration, search, feature selection or narrative-driven reruns. 2025-26 is reserved for a future external extension; no outcome analysis of that season in this release.

Models and fixed search spaces

1. Expanding home/away league Poisson baseline, updated from strictly earlier dates.

2. Smoothed 20-game team-form Poisson baseline: average attack and opponent defence.

3. Ridge point prediction embedded in an independent Poisson distribution (a proxy, not a fitted count likelihood): penalties 1, 10 and 100.

4. Penalized independent Poisson log-link model: penalties 0.01, 0.1 and 1.

5. NB2 log-link model: penalties 0.01 and 0.1; training-estimated dispersion.

6. Penalized attack/defence Poisson: team and opponent indicators, shared home effect, league rate, penalties 0.01, 0.1 and 1; unseen teams receive zero categorical effect. This is a frequentist shrinkage model, not Bayesian posterior inference.

7. Plug-in bivariate Poisson: same marginal Poisson feature model and search; training-estimated shared component c=rho*min(mu_home,mu_away), 0<=rho<=0.8. This allows only positive conditional dependence. It is not jointly optimized bivariate regression and does not guarantee a gain.

8. Histogram gradient boosting with Poisson loss: 7/15 leaves crossed with 100/200 iterations, learning rate 0.05, minimum leaf 50, L2 penalty 1, no random internal early-stopping split.

9. Random forest point means embedded in Poisson margins: minimum leaf 20/50, 250 trees, max features 0.8; this distribution is also explicitly a proxy.

Neural networks, Bayesian sampling, roster/goaltender information, xG, dynamics, Dixon-Coles correction and calibrated ensembles are outside this first experiment. They remain future hypotheses. They are not implied by completed model names.

Evaluation and selection

Primary: paired-score mean negative log likelihood, lower is better. Secondary: side and combined MAE/RMSE, bias, totals and differential MAE, marginal central 90% interval coverage and width, regulation home/draw/away Brier and log loss. For discrete counts, equal-tail 90% intervals can cover more than 90%. Outcome probabilities use a 0..35 score grid; reject material truncated mass. Joint log likelihood is evaluated on the untruncated distribution.

Compare all models on the same games. Use 2000 paired calendar-week bootstrap replicates for NLL differences; acknowledge cross-week team dependence and the limited single-holdout horizon. Report every model, fold and exclusion, including negative results. Draw calibration uses fixed probability bins. Do not claim causality, profit, institutional acceptance or that any hypothesis is proven.

Completion and release

Version 0.1.0 can release the source-verified audit, this executable first experiment, formal PDF/DOCX documents and reproducibility tooling. Version 1.0.0 requires archival status resolution, independent review, full sensitivity work, source-time/score verification, R parity and clean-environment reproducibility.

Appendix A Machine readable data dictionary

The canonical table preserves source game ID, season start year, source date, home and away team labels, regulation goals, actual played goals, duration, overtime/shootout indicators, and prior-game shot/attempt totals. The analytical feature table adds strictly earlier-date histories.

date is a source calendar-date boundary, not a verified UTC publication timestamp. history_date is the latest source date admitted for the team. The source-date invariant requires history_date < date. A missing first history is represented as no prior observation, with a fixed pooled prior.

Field or family	Definition	Availability or policy
game_id and season	Stable source game ID; starting-year season	One unique paired row
home_goals and away_goals	Reconstructed regulation counts	Outcome only; never a current predictor
duration	All-situation elapsed seconds	Target reconstruction only
played_goals	Actual goals including overtime	Exclude shootout scoreboard marker
team_gf and team_ga	Smoothed prior 20-game regulation goals	Earlier source dates only
team_sf and team_sa	Smoothed prior-game shots	Overtime exposure may remain
team_cf and opp_ca	Smoothed prior-game shot attempts	Explicit lag; correlated predictors regularized
rest and opp_rest	Days since prior included game; cap 7	Calendar-date proxy
league_rate	Expanding side-specific earlier-date mean	100 fixed prior games
team and opponent	Categorical team labels	Unseen category pooled in model

Appendix B Model and information cards

Model scope: historical NHL regular-season paired regulation counts. Intended use: non-commercial comparative research and education. Excluded use claims: guaranteed exact scores, causal intervention effects, betting profitability, final winner probabilities without an overtime model, or institutional acceptance.

Primary configuration: the frozen bivariate-Poisson label, with estimated rho zero, produces the independent-Poisson distribution. Independent and bivariate variants are numerically tied. A future tie-selection rule should explicitly prefer the simpler model, but no post-holdout relabeling changes this release’s frozen record.

Information set: historical goals, shots, attempts, source-date rest and home role. Roster, injuries, confirmed goalies, odds, xG, player tracking, travel and exact venue context are not part of the primary model. Their absence must not be interpreted as proof that they lack value.

Known limitations: revised retrospective source snapshots; calendar-time proxy; sample-based score verification; one excluded incomplete game; limited final-season horizon; positive-only shared-count dependence; discrete interval overcoverage; proxy distributions for squared-error estimators.

Appendix C Frozen configurations and environment

Model	Frozen final choice
Ridge Poisson proxy	{"alpha": 100.0}
Independent Poisson	{"alpha": 1.0}
Negative binomial	{"alpha": 0.1}
Attack and defence Poisson	{"alpha": 0.01}
Bivariate Poisson	{"alpha": 1.0}
Poisson boosting	{"iterations": 100, "leaves": 7}
Forest Poisson proxy	{"leaf": 50}

Python interpreter used for fitting: 3.12.14. Random seed: 20261002. Training is deterministic under the stated software and numerical tolerance, with four computation threads used for reproducible bounded parallelism.

Package	Recorded version
numpy	2.3.5
pandas	3.0.1
scipy	1.18.1
scikit-learn	1.9.1
matplotlib	3.11.2

R parity uses base R 4.2.3 and no contributed packages. Its interpreter and session record define that environment. A broader contributed-package R suite would require its own dependency lock. Python dependency pins and a generated full dependency lock document the actual environment.

Appendix D Verification and release gates

Fast verification checks count support, score reconstruction edge cases, future-outcome mutation invariance, same-date batching, side stacking, bivariate mass/moments, outcome mass and bootstrap pairing. CI can execute these on synthetic data without network access or third-party raw downloads.

Full replay validates the exact raw and protocol hashes, refits only frozen configurations and requires aggregate holdout metrics to agree within 1e-7. It does not select hyperparameters again. The R Poisson parity threshold is 1e-6.

Publication retains the original PDF, provenance hashes, aggregate tables, model choices, protocol, tests, manuscript sources and final documents. Raw and row-level outputs remain excluded. Software licensing does not confer rights over MoneyPuck or NHL content.

A stable 1.0.0 research release still requires resolved archival status, independent review, broader score and availability verification, sensitivity analysis and any justified advanced models. The current 0.1.0 release must keep that remaining work visible.

References

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